Behavioral Reconstruction Module - (BRM)

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Independent Methodological Authority

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Canonical Definition System

Canonical Definition

Behavioral Reconstruction Module (BRM) defines the structural conditions under which AI behavior can be reconstructed from preserved behavioral records, operational context, governance evidence, decision pathways, interaction history, system states, and resulting behavioral outcomes throughout the complete behavioral lifecycle.

BRM governs behavioral reconstruction.

The module establishes the foundational conditions required to ensure that significant AI behavior can be reconstructed after execution for governance review, regulatory oversight, operational investigation, accountability analysis, and trust validation.

AI behavior may occur in real time.

Governance must be able to reconstruct it later.

BRM governs that reconstruction.

Module Operational Role

BRM defines the behavioral-reconstruction layer of BAS.

Its role is to preserve the ability to reconstruct AI behavior after execution.

Module Operational Space

BRM governs:

- behavioral reconstruction
- behavioral history reconstruction
- decision-pathway reconstruction
- interaction reconstruction
- operational context reconstruction
- governance evidence linkage
- behavioral lifecycle reconstruction
- audit reconstruction

The module applies wherever AI behavior must be reconstructed for governance, audit, compliance, investigation, or accountability purposes.

Module Function

The module applies wherever systems must preserve:

- reconstructable AI behavior
- traceable behavioral history
- linked operational context
- governance-valid behavioral records
- auditable decision pathways
- operational behavioral evidence

Its function is to ensure that significant AI behavior can be reconstructed from available governance evidence rather than inferred from incomplete assumptions.

Minimum Implementation Framework

1. Define the Behavioral Reconstruction Object

The organization must define which AI behaviors require reconstruction capability.
This may include:

- autonomous decisions
- conversational interactions
- recommendations
- robotic actions

- escalation events
- Human-AI decisions
- multi-agent actions
- safety-critical behaviors

2. Define Reconstruction Conditions

The system must define the conditions required to reconstruct behavior.
This includes:

- behavioral records
- system-state information
- input context
- output history
- governance evidence
- decision-pathway data
- intervention records

3. Define Reconstruction Degradation Detection Logic

The system must define how reconstruction failure is identified.
This may include:

- missing behavioral history
- incomplete context
- broken evidence links
- unreconstructable decisions
- missing interaction records
- governance-invalid reconstruction conditions

4. Define Operational Response or Governance Logic

The system must define governance logic for behavioral-reconstruction failures.
Governance response may include:

- reconstruction review
- evidence recovery
- behavioral investigation
- governance intervention
- audit escalation
- corrective documentation
- operational invalidation where required

5. Preserve Traceability & Restrict Invalid Conditions

The system must preserve reconstructable traceability of:

- behavioral records

- decision pathways
- operational contexts
- governance reviews
- evidence links
- resulting behavioral states

An AI behavioral environment must not remain auditability-valid if materially significant behavior cannot be reconstructed, reviewed, validated, preserved, or governed.

Use Case 1 — AI Loan Decision System

Scenario

An AI system recommends loan approval or rejection based on financial data, risk indicators, and internal policy rules.

Application

BRM reconstructs the behavioral pathway from input data through recommendation logic, governance evidence, and final behavioral outcome.

Result

The financial institution gains stronger auditability, improved regulatory readiness, clearer accountability, and reduced governance uncertainty.

Use Case 2 — Autonomous Industrial Robot

Scenario

An autonomous robot performs an unexpected operational action during a manufacturing process.

Application

BRM reconstructs the behavior from sensor data, operational context, system state, behavioral logs, and governance evidence.

Result

The organization gains stronger incident investigation capability, improved operational safety, and better behavioral governance.

Canonical Closing Statement

Behavioral Reconstruction Module (BRM) defines the structural conditions under which AI behavior can be reconstructed from preserved behavioral records, operational context, governance evidence, decision pathways, interaction history, system states, and resulting behavioral outcomes throughout the complete behavioral lifecycle.

Behavior that cannot be reconstructed cannot be independently audited.
Behavioral Reconstruction therefore becomes the first operational condition of Behavioral Auditability Standard within AI Governance Architecture (AIG®).

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Canonical Source

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